lines of (20c), her utterance will achieve some cognitive effects in the hearer's mind, but she can not foresee of what nature they may be. In such a situation, it is actually not possible for Mary to convey any implicatures, and her utterance is relevant to the audience in virtue of its explicature alone.

Since relevance consists of the balance of cognitive effects and processing effort, relevance expectations may relate to these factors individually or jointly: an audience may at times have rather specific expectations about the type or content of cognitive effects that would be relevant to him. At other times, he may have more general expectations about the amount of cognitive effects to be achieved. On yet other occassions, the audience's relevance expectations may relate to the effort side. Sperber and Wilson (1995, p. 237) discuss the example of Gustave Flaubert commenting on the poet Leconte de Lisle:

(21) His ink is pale. (Son encre est pale.) [Sperber and Wilson's example 108]

The audience's expectations of relevance before processing this utterance are specific enough in the sense that it expects the utterance to be relevant by way of giving some indication of Flaubert's opinion on de Lisle's poetry. But there is no strong indication of what direction these indications may take. In other words, the audience is processing the utterance with the expectation that its processing effort will be offset by enough cognitive effects to be worth the effort (if not more), but it has no prior expectations of which type of content they may have. Sperber and Wilson's analysis is worth quoting in full:

> A strictly literal construal of this utterance is clearly ruled out: it is hard to see what relevance could attach to knowing the colour of a poet's ink. Nor is there any obvious strong implicature. The only way of establishing the relevance of this utterance is to look for a wide range of very weak implicatures. This requires several extensions of the context. In the most accessible context of encyclopaedic information about ink and handwriting, most implications are irrelevant: after all, Leconte de Lisle's poetry is read not in his handwriting but in print; the only clear implicature in this first context is that he has the character of a man who would use pale ink. Some other implications – that Leconte de Lisle's writing lacks contrasts, that it may fade – have further relevant implications in a context to which has been added the premise that what is true of his handwriting is true of his style. Someone who knows little of Leconte de Lisle's work might conclude, for example, that there is something weak about his poetry, that his writings will not last, that he does not put his whole heart into his work, and so on. Someone who has a deeper acquaintance with the poet would be able to construe the criticism in much more detailed and pointed ways. The resulting interpretation, with its characteristic poetic effect, owes a lot simultanewouly to Flaubert, for foreseeing how it might go, and to the reader, for actually constructing it. (Sperber and Wilson 1995, p. 237)

The upshot of this is that the composition of the content of the communicator's informative intention – the speaker's meaning – varies in relation to the nature of the extant relevance expectations in the communication situation. Strong implicatures can only be communicated in situations where the relevance expectations are strong

Relevance expectations	Speaker's meaning	Example
strong, specific	strong implicatures, explicature(s) and some weak implicatures	(1b)
strong, non-specific	weak implicatures and explicature(s)	(21)
weak, non-specific	explicature	(20)

Table 2: The nature of relevance expectations and the composition of the content of the speaker's informative intention

and specific enough to guide the audience in backwards inference. As we have seen, some weak implicatures are communicated in this case as well to give a rational for speaking implicitly at all. When the relevance expectations are non-specific and weak, the communicator can not convey implicatures, and the utterance will be relevant in terms of its explicature alone. When relevance expectations are strong but non-specific, the speaker's meaning may consist of a fuzzy set of lots of weak implicatures (and presumably explicatures giving access to these implicatures as well). This is summarised in table 2. Needless to say that these categories are not exclusive ones; rather, many utterances in verbal communication will fall somewhere in between the space plotted out by these examples, with respect to the strenght and specifity of the relevance expectations raised and the composition of the content of the speaker's informative intention.[9]

4.3.2 Relevance and the utility of utterances

Rational speech act theory models speakers as Bayesian decision makers who choose their utterance optimizing its expected *utility* to get the audience to comprehend the speaker's meaning. Much previous work in RSA has focussed on communication situations where the utterance is relevant solely in virtue of its explicature: reference games (Frank and Goodman 2012) or the semantics and pragmatics of scalar expressions[10] (Goodman and Stuhlmüller 2013)[11]. In these instances, the utility of the utterance was seen as a reflection of its power to guide the audience to an informative interpretation, and this in turn was measured in the expression's information-theoretic *negative surprisal*. However, in most communication events the speaker's meaning is not comprised of one proposition alone, but several: an explicature and several implicatures, each of which the communicator intends to make more manifest to the audience to varying degrees. How should the utility function be defined in this case?

Consider again what it means to say that Mary communicates the explicature in table 1 (e) and the implicatures in table 1 (d), (f), (g), (h) to Peter. According to the definition of the communicator's informative intention in (5a), it means that Mary's utterance

[9]Recall furthermore that communication does not aim at achieving identity of the audience's thoughts and speaker meaning; rather, communicators aim at a relevant overlap of thoughts.

[10]These are usually analysed as carrying *scalar implicatures* (Horn 1984), but see R. Carston (1995) for arguments that the pragmatic contributions of scalar terms affect explicatures rather than implicatures.

[11]See Bergen, Levy and Goodman (2016) for a noticeable exception to the trend of focussing narrowly on scalar inferences. Bergen, Levy and Goodman study so-called M-implicatures, where the speaker uses a more elaborate, more expensive to process way of expression than necessary (imagine Mary answering Peter: *I do not ingest melatonin receptor blockers less than 6 hours before my regular bed time*, which would not only allow Peter to infer that Mary does not want coffee now, but more likely than not convey more implications such as that Mary may be a nerd that is difficult to deal with). Still, this type of implicatures is only a subset of conversational implicatures, and so the question arises how well their model might generalise to the full range of implicatures.

alters the manifestness of each of these propositions for Peter. This is paramount to saying that Mary changes Peter's probability distribution over these propositions from a prior value to a posterior value. The utility of Mary's utterance can then be measured in terms of the divergence between Peter's prior probability distribution over the set of these propositions to Peter's posterior probability distribution over this set. In information-theoretic terms, this is the Kullback-Leibler divergence between the two probability distributions in question.[12]

There is an other way to conceptualise the utility of an utterance. The utility of an utterance can be understood as the efficiency with which it aligns the audience's interpretation with the communicator's informative intention, in other words: its relevance (in the relevance-theoretic sense). Would it then not be simpler to directly calculate with estimates of cognitive effects vs. processing effort measures? For example, could we not make rough estimates about the number of cognitive effects and approximate the processing effort to the number of inference steps and/or context extension steps? Such estimates would require us to make significant assumptions about actual cognitive implementation which are most likely wrong. In addressing these matters in some detail, Sperber and Wilson (1995, pp. 123–132) conclude that relevance is a non-representational property:

> Mental effects and effort are non-representational properties of mental processes. Relevance, which is a function of effect and effort, is a non-representational property too. That is, relevance is a property which need not be represented, let alone computed, in order to be achieved. When it is represented, it is represented in terms of comparative judgements and gross absolute judgements, (e.g. 'irrelevant', 'weakly relevant', 'very relevant'), but not in terms of fine absolute judgements, i.e. quantitative ones. (Sperber and Wilson 1995, p. 132)

We submit that the Kullback-Leibler divergence between the audience's prior and posterior probability distributions over the set of communicated assumptions is a viable way to conceptualise comparative judgements of relevance in an indirect way, which is suitable for developing probabilistic models of utterance interpretation while avoiding the trap of trying to quantify non-representational properties.

4.3.3 Relevance expectations as a stopping criterion for the heuristic procedure

A cognitive heuristic procedure is triggered by the recognition of an input stimulus affecting the domain of the procedure. In the case of communicative stimuli, this will mean stimuli that make a layered set of communicative and informative intentions overt. The procedure must also stop at some point, so a heuristic procedure must also specify a stopping criterion. In the case of communication, the stopping criterion is the satisfaction of the audience's expectations of relevance *effective in the present communication situation.*

We have already seen in section 4.3.1 that relevance expectations may be of different types according to the different factors of cognitive effects and processing effort that make up relevance:

[12]Tessler, Tenenbaum and Goodman (2022, p. 583) deal with a similar problem when they consider a speaker communicating her conclusion of a syllogistic argument to an audience. The conclusion of a syllogistic argument is comprised of a set of states, and the utility of a conclusion can be "quantified by how well it would align the listener's beliefs with those of the reaoner (speaker)."